12.47 MME 386 Marine Electrical Technology and Electrical Measurements (2, 0, 2)
Course objectives:

1. Teach on electrical propulsion system.
2. Teach on how Maintenance of electrical equipment is done.
3. Explain Basic requirements of a measuring instrument.
4. Give the students classification of requirements for electrical equipment for steering gears, electrical control from wheel house for electro-hydraulic steering gear.

Content:

1. Essential equipment and special circuits- Classification of requirements for electrical equipment for steering gears, electrical control from wheel house for electro-hydraulic steering gear, all electrical steering gear circuit and control equipment, navigation lights. Miscellaneous marine equipment and alarm system-electrical engine room telegraph, remote indicating revolution counter, remote helm indicator, salinity indicators, mist detectors, carbon di-oxide recorder, electrical equipment for water tight door operation, centralized alarm system in machinery space, for the above, fire alarm system for engine room, holds and accommodation, high temperature alarms low and high level alarm.

2. Electric propulsion –systems, difference and relative merits, alternating current, diesel electric and turbo electric propulsion system engines, excitation system and control.

3. Maintenance of electrical equipment-detection of faults and repair, preventive maintenance and periodic survey of equipment. Localization of cable faults. Classification of requirements for spares on board.

4. Basic requirements of a measuring instrument-deflection, control and damping devices moving coil, moving iron, dynamometer and thermocouple type of ammeter, voltmeter and wattmeter-their construction and other details, extension of scales of a meter. Single phase and three phase measurements by wattmeter. Measurement of energy. Measurement of speed, frequency and phase difference. Measurement of resistance, inductance and capacitance by Bridge Method. Magnetic measurement. Transducers and its application in the measurement of pressure, flow, temperature etc. Simple electronics measuring devices, such as VTVM, CRO, IC tester, signal generator.

Reading material:

Design Principles of Ships and Marine Structures Suresh Chandra Misra November 24, 2015 by CRC Press ISBN 9781482254464